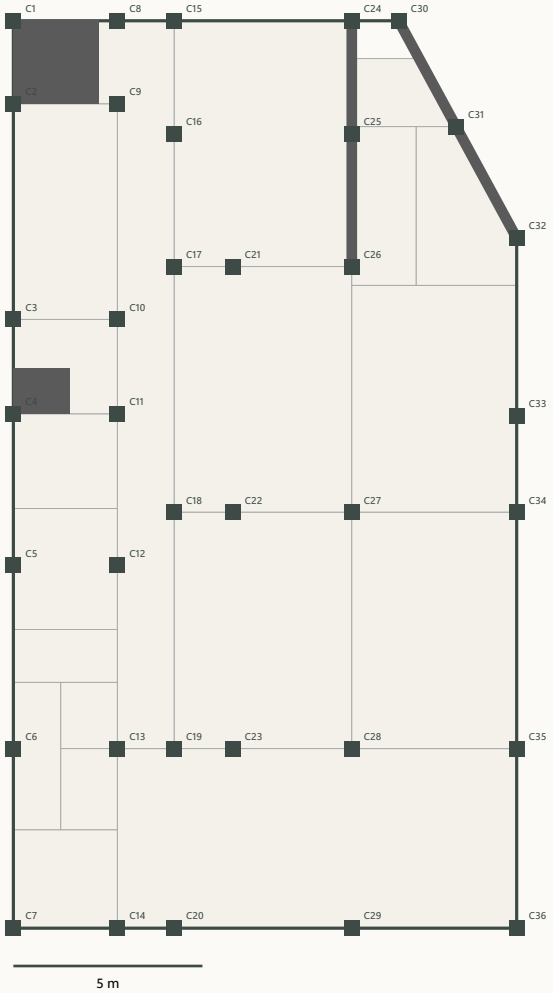
Preliminary structural study · Taif / Palm House V6.3

architectural coordination study · every size ASSUMED, to be calculated by the structural engineer · beam depths are TOTAL depths including the slab · CSV coordinates, north up

ALL CHECKS PASSED means the drawings are consistent; it does not mean strength, deflection or vibration were calculated.

Ground floor · columns (■ in the walls, ● colonnade, all to the roof: X8), RC boxes (dark), tie beams on the column lines



The idea: a simple RC skeleton, walls on walls

- A **reinforced-concrete frame** (columns, beams, slabs) with **block infill**, as Saudi contractors build every villa. Columns stand on the existing wall lines, hidden in the walls; **every column runs from its footing to the roof** (the three colonnade columns too, since X8: they carry the edge beam B3 and the roof edge beam R11). No transfer beam carries a column.
- **Spans 2.2–6.5 m.** The only spans above 6 m are on the courtyard lines (x 4.25, 5.8, 8.95: 6.5 and 6.25 m), because the courtyard and the double-height living must stay free of columns; they get drop beams ≈ 0.50–0.55 total depth (0.30–0.35 below the slab).
- **Slabs:** solid 0.20 m where spans are small (west band, riwaq, projection, roofs of 4.7 m), hollow-block **ribbed 0.30 m** (25+5) for the 4.36–4.75 m bays of the east band and the men's majlis roof. The double-height living's roof is a **flat solid slab 0.22 m spanning 4.7 m east–west** between the two long walls: **no beam crosses the void** (§ options; a common span for a 0.22 solid slab, not calculated).
- **Foundations (assumed):** isolated pad footings ≈ 1.5 × 1.5 m under the columns, strip footings under the RC boxes, RC tie beams on the column lines at ground level; the lift pit 1.2 m deep; the air duct (studies/air_system.typ v2, route A): a small dry sealed box 0.9 × 0.9 (0.6 × 0.6 clear, assumed) leaving the tower base pit westward and running under the women's majlis to a chamber under the courtyard floor: top −1.00, just under the tie beams, clear of the pads except at its two ends. **No soil report yet** (X9).

Balanced stiffness · a proposal, NOT calculated

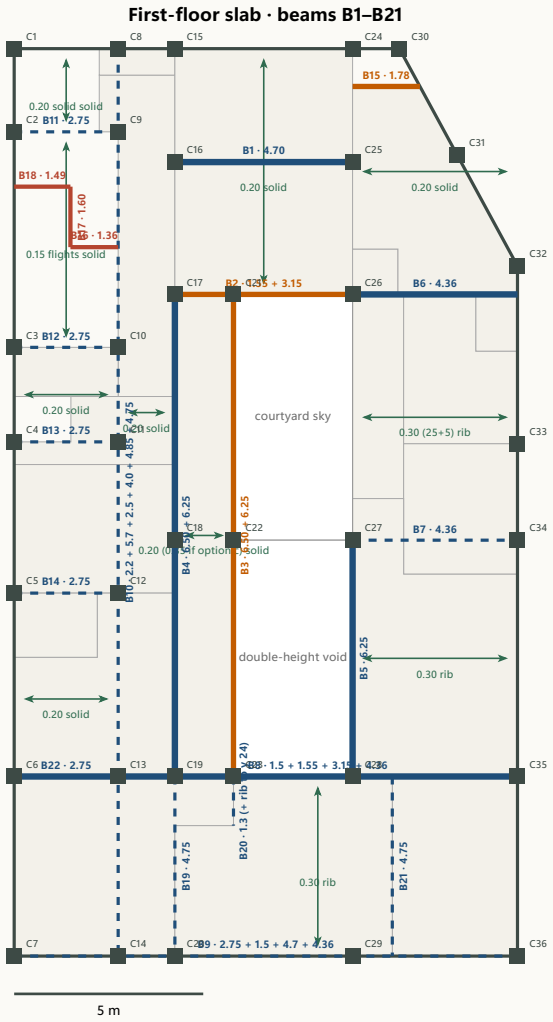
- **West:** the lift shaft (W1, RC walls on 3 sides), the west shaft (W2) and the stair walls make a stiff core at x 0–2.75 (its stiffness is not calculated here).
- **East, to match it:** the solid laundry / maid wall on x 8.95 (W3, no openings, 6.5 m) and the cut wall (W4) are proposed as RC walls 0.20 m (W5, the kitchen's north wall, is dropped in V6.4: that wall is now a partition at y 7.0 and the y 6.5 line is the beam B6), intended as an east box of a similar order of stiffness at a similar distance from the centre, so that the centre of stiffness stays near the centre of mass. **Not calculated:** the engineer models the lateral system (frame + walls) and confirms it.
- **Why:** the house is a long pentagon (13.3 × 24 m); with stiff elements only on the west, wind and earthquake would tend to twist it about the west core. The seismic design category for Taif and the wind case are to be taken from SBC 301 by the engineer; not assessed here.

Column lines and spans

Line	Columns at y (or x)	Spans (m)
x 0 and x 2.75	y 0 · 2.2 · 7.9 · 10.4 · 14.4 · 19.25 · 24	2.2 · 5.7 · 2.5 · 4.0 · 4.85 · 4.75
x 4.25 and x 8.95	y 0 · 3.0 · 6.5 · 13.0 · 19.25 · 24	3.0 · 3.5 · 6.5 · 6.25 · 4.75
x 5.8 (colonnade, to the FF)	y 6.5 · 13.0 · 19.25	6.5 · 6.25
cut wall	(10.188, 0) · (11.709, 2.8) · (13.311, 5.749)	3.2 · 3.6
x 13.311	y 6.0 · 10.45 · 13.0 · 19.25 · 24	4.45 · 2.55 · 6.25 · 4.75
y 0 / y 24 (cross)	x 0 · 2.75 · 4.25 · 8.95 · 13.311 (cut corner at 10.188)	2.75 · 1.5 · 4.7 · 4.36 (1.1)
y 6.5 · 13.0 · 19.25 (cross)	x 4.25 · 5.8 · 8.95 · 13.311 (+ 2.75 at y 19.25)	1.55 · 3.15 · 4.36

First-floor slab (+3.40 … +3.60) · beams and slab directions

blue = drop beams · dashed = hidden beams (slab depth) · orange = edge beams of the openings · red = stair trimmers · green arrows = slab span · sizes width × depth, assumed



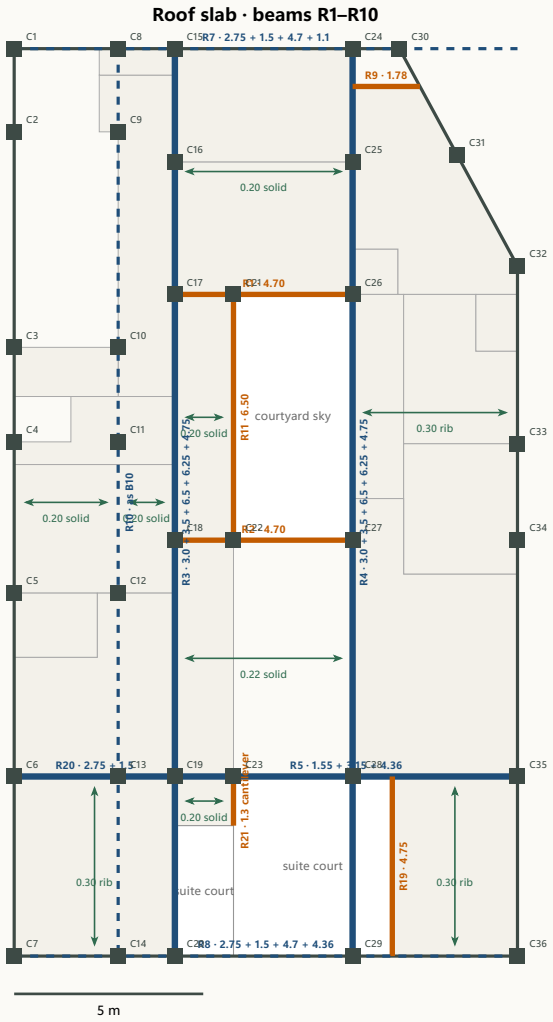
Beam	Span (m)	Size, assumed	Carries / note
B1	4.70	0.25×0.40 total = 0.20 below the slab, soffit +3.20; hidden by the women's majlis gypsum ceiling (≈ +3.05, owner decision)	FF kids' room / family living wall (y 3.0, nothing below) + the women's majlis roof
B2	1.55 + 3.15	0.25×0.40 hidden / drop	FF slab edge on the courtyard (the family living's steel window W2 stands on it, V6.4); over the women's majlis south glazing
B3	6.50 + 6.25	0.25×0.50 total · UPTURNED (owner's preference: 0.30 above the slab = the sill of the gallery window W1, V6.4) or drop (0.30 below; the engineer decides) · § options	the 1.35 m projection and the gallery, on C21–C23
B4	6.50 + 6.25	0.25×0.50 total = 0.30 below the slab, soffit +3.10	FF west rooms' east wall (over the riwaq glazing and the living's entrance arch; on the solid TV wall from y 14.5, V6.3) + the projection back span
B5	6.25	0.25×0.55 total = 0.35 below the slab, soffit +3.05 (or 0.60×0.40 wide-shallow, soffit +3.20)	FF suite bedroom west wall + half the dining roof; over the open ceiling-height change
B6	4.36	0.25×0.45 total = 0.25 below the slab, soffit +3.15 (V6.4; was hidden 0.30 in the wall)	V6.4: no wall below any more (the kitchen's north wall moved to y 7.0 as a non-load-bearing partition): a drop beam from C26 to the east wall inside the laundry / maid room ceiling, hidden by the gypsum ceiling (≈ +3.0); carries the FF bedroom 1 / suite wall above
B7	4.36	hidden 0.30	on the kitchen / dining wall; the suite dressing floor spans over it
B8	1.5 + 1.55 + 3.15 + 4.36	0.25×0.45 total, 0.25 above the wall	on the men's majlis north wall; carries the FF double wall bedroom 3 / gym (x 2.75–4.25), the court's north edge (the living's south wall) and the suite bedroom / office wall (v2)
B9	2.75 + 1.5 + 4.7 + 4.36	hidden in the 0.25 wall / 0.25×0.45	south wall; men's majlis roof support
B10	2.2 + 5.7 + 2.5 + 4.0 + 4.85 + 4.75	hidden 0.20–0.30; 0.25×0.40 at the 5.7 m stair bay	riwaq line: west band slabs and the riwaq slab
B11	2.75	hidden 0.20	lift / stair wall
B12	2.75	hidden 0.20	stair / utility wall · FF landing
B13	2.75	hidden 0.20	utility / hospitality kitchen wall
B14	2.75	hidden 0.20	basins / family WC wall · FF bedroom 2 / bath 3 wall
B15	1.78	0.20×0.35 edge	air tower opening (the tower rises through both slabs)
B16	1.36	0.20×0.40 total, soffit +3.20	stair flip (2 Nov 2026, B16'): the opening edge over the lower (east) flight beside tread 6 (headroom 2.22 m to the beam soffit); from the riwaq line beam B10 to B17' (beam on beam: normal at a stair well)
B17	1.60	0.20×0.40 total, soffit +3.20	B17': on the lower flight's side of the split (x 1.39–1.59), never over the upper (west) flight; ends on B16' and B18'
B18	1.49	0.20×0.40 total, soffit +3.20	B18': the arrival edge (top riser line y 3.75) under the arrival strip, from the west wall to B17'
B19	4.75	0.60×0.30 hidden (in the ribbed slab depth) or 0.25×0.50 drop, between C19 and C20	v2: the gym's east wall = the court's west wall, on a beam between two existing columns (a frame line, no transfer); its roof edge R3 above
B20	1.3 (+ rib to y 24)	hidden rib 0.30	v2: the gym nib's east wall from C23, a light partition with the court door; nothing below
B21	4.75	hidden rib 0.30	v2: the court / office wall x 10.0, NON-load-bearing (the office roof spans N–S onto R5 / R8): its own weight only on a thickened rib
B22	2.75	0.25×0.45 total, drop 0.25 below the 0.20 slab (soffit +3.12)	v2: the double wall bedroom 3 / gym over the guest passage and the guest WC door (no GF wall below on x 0–1.25), between C6 and C13

First-floor walls and what carries them

On a wall below (block infill under a hidden beam)	Nothing below → beam or light partition
x 0, x 13.311, y 0, y 24, the cut wall (outer walls); x 2.75 at the lift and landing; x 2.25 lift; y 2.2, 7.9, 10.4 (shaft), 14.4, 16.1 (bath 3 = GF family WC), 19.25 (majlis north wall), x 1.5 shaft; x 8.95 y 6.5–13 (courtyard east wall); y 13.0 line inside the suite (kitchen / dining wall)	**y 3.0** kids' / family living → **B1**; **x 4.25 y 13–19.25** bedrooms' east wall over the riwaq glazing and the entrance arch (on the solid TV wall from y 14.5) → **B4**; **x 8.95 y 13–19.25** suite bedroom west wall over the open ceiling change → **B5**; **y 6.5 x 8.95–13.311** bedroom 1 / suite wall over the maid wing (the kitchen's north wall moved to y 7.0, V6.4) → **B6** drop beam; **x 4.25 y 19.25–24** office → **B19** rib; **x 5.6** gym route → **B20**; **x 11.586** treadmill arm → **B21**; **y 21.25** gym north wall (7.7 m, east–west across the ribs) → **light partition on the slab** (X6); suite walls x 10.3, y 10.45, x 12.211, y 8.0, y 11.9, y 13.9 over the kitchen → light partitions (100 mm block or dry wall) on the ribbed slab, or hidden beams under the bath walls; y 11.0 (bath / niche / bedroom 2) and y 9.2 x 1.5–4.25 (family bath north) → light partitions; x 2.2 bath 3 / bedroom 3 entry → light; y 5.3 and x 10.15 (bedroom 1 / family link) → light; linen front y 0.7 → joinery

Roof slab (+7.00 … +7.20) · beams and slab directions

the courtyard sky and the suite court are holes; the double-height living is roofed by a flat slab with no beam in the void; the colonnade C21–C23 continues to the roof (X8), the gallery strip roof on R11



Beam	Span (m)	Size, assumed	Carries / note
R1	4.70	0.25×0.45 drop	courtyard sky hole, north edge (over the FF family living balustrade line)
R2	4.70	0.25×0.45 drop	courtyard sky hole, south edge = the clean double-height roof's north edge
R3	3.0 + 3.5 + 6.5 + 6.25 + 4.75	0.25×0.50 total, 0.30 above the FF wall	on the FF west rooms' east wall; the gallery strip roof spans 1.35 m from it to R11 (X8); v2: on to y 24 over the gym's east wall = the court hole's west edge
R11	6.50	0.25×0.45 total, soffit +6.75	roof edge of the courtyard sky over the gallery strip, on C21–C22 (X8, owner: the colonnade continues to the roof, no cantilever)
R4	3.0 + 3.5 + 6.5 + 6.25 + 4.75	0.25×0.50 total, 0.30 above the FF wall	on the FF suite west wall; the double-height roof spans to it
R5	1.55 + 3.15 + 4.36	0.25×0.50 total, soffit +6.70	over the living's south wall (clerestory head +6.00, 0.70 m below the soffit) and the suite court north wall
R19	4.75	edge rib 0.30 (thickened), on R5 and R8	v2: the office roof's west edge = the court hole's east edge; the office roof spans N–S, so the x 10.0 wall below carries nothing (R6 of V7.1 / v1 removed)
R20	2.75 + 1.5	0.25×0.45 drop	v2: over the double wall bedroom 3 / gym: the gym roof's north support (C6 – C13 – C19)
R21	1.3 cantilever	0.20×0.40	v2: the gym nib's roof edge, a cantilever from C23 / R5 (or a light steel post at the corner): engineer
R7	2.75 + 1.5 + 4.7 + 1.1	hidden in the 0.25 wall	north wall
R8	2.75 + 1.5 + 4.7 + 4.36	hidden in the 0.25 wall; 0.25×0.60 over the court (C20 – C29 – C36)	south wall; v2: across the court's south side it is the facade frame carrying the cornice and the parapet over the open court (the infill below is the owner's choice)
R9	1.78	0.20×0.35 edge	air tower head opening
R10	as B10	hidden 0.20–0.30	riwaq line

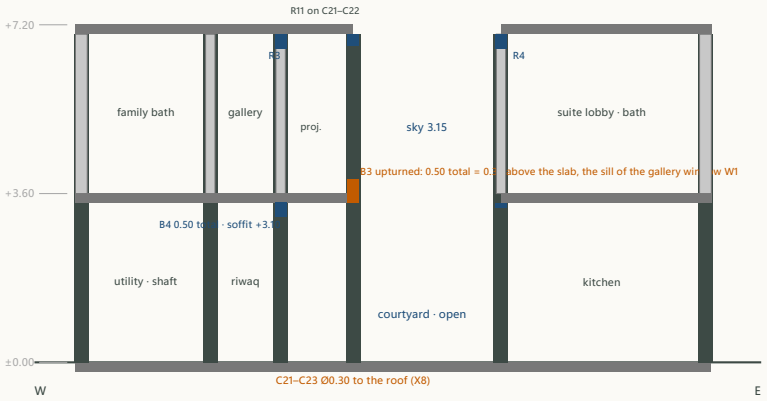
Slab bays (first floor and roof)

Level	Bay (x0 y0 – x1 y1)	Type	Span	Note
FF	0 0 – 2.75 2.2	lift: FF slab opening (the lift is installed at construction, V6.3) 0.20 solid	NS	lift
FF	0 2.2 – 2.75 7.9	stair: L-shaped opening + flight slabs 0.15 flights	NS	flipped (2 Nov 2026): opening x 0.125–1.39 y 3.75–7.80 + x 1.39–2.65 y 5.35–7.80 (about 8.2 m2), the arrival strip y 2.30–3.75 and the slab over the lower flight's treads 1–6 at the north end, see the stair page
FF	0 7.9 – 2.75 10.4	solid 0.20	EW	utility; shaft opening 1.5×1.2 framed
FF	0 10.4 – 2.75 24	solid 0.20	EW	west wash block, 2.75 m bays
FF	2.75 0 – 4.25 19.25	solid 0.20	EW	riwaq, 1.5 m
FF	4.25 0 – 8.95 6.5	solid 0.20	NS	women's majlis roof: 3.0 + 3.5 m to beam B1
FF	4.25 6.5 – 5.8 19.25	solid, the projection 0.20 (0.35 if option c)	EW	1.55 m between B4 and the edge beam B3 (V6.4)
FF	8.95 0 – 13.31 6.5	solid 0.20	EW	maid wing, 4.36 m to the cut wall
FF	8.95 6.5 – 13.31 13	ribbed (hollow block) 0.30 (25+5)	EW	kitchen roof; suite bath sunken 0.15 (depressed zone x 10.3–13.311 y 6.5–10.45)
FF	8.95 13 – 13.31 19.25	ribbed 0.30	EW	dining roof; half on beam B5
FF	2.75 19.25 – 13.31 24	ribbed 0.30	NS	men's majlis roof, 4.75 m; thickened ribs B19–B21 under the FF walls; gym live load
roof	0 0 – 2.75 24	solid 0.20	EW	west band and lift / shaft heads
roof	2.75 0 – 4.25 24	solid 0.20	EW	riwaq / gallery line
roof	4.25 0 – 8.95 6.5	solid 0.20	EW	kids' room / family living: 4.7 m, no beam on y 3.0 (light partition)
roof	4.25 6.5 – 5.8 13	solid 0.20	EW	over the gallery strip: 1.55 m from R3 to the roof edge beam R11 on C21–C22 (X8; V6.4)
roof	4.25 13 – 8.95 19.25	solid, flat, no beams in the void 0.22	EW	the double-height living roof: 4.7 m R3 → R4, clean soffit
roof	8.95 0 – 13.31 19.25	ribbed 0.30	EW	bedroom 1 / suite: 4.36 m
roof	0 19.25 – 4.25 24	ribbed 0.30	NS	v2: the gym roof, 4.75 m from R20 to R8 (light-roof option: an insulated steel deck, to decide)
roof	4.25 19.25 – 5.8 20.55	solid 0.20	EW	v2: the gym nib's roof, 1.55 m from R3 to R21
roof	10 19.25 – 13.31 24	ribbed 0.30	NS	v2: the office roof, 4.75 m from R5 to R8, its west edge rib R19 on the court (light-roof option)

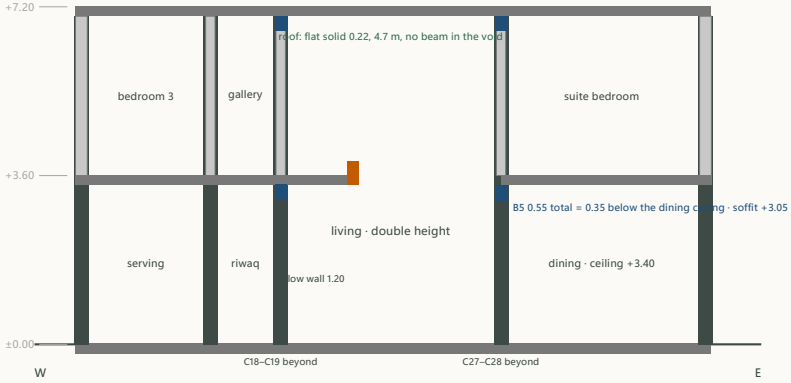
Sections · schematic, levels to confirm

A east–west through the riwaq, the courtyard and the colonnade (y 10) · B east–west through the living, the double-height void and the dining (y 16.5) · C north–south through the middle band (x 7.3)

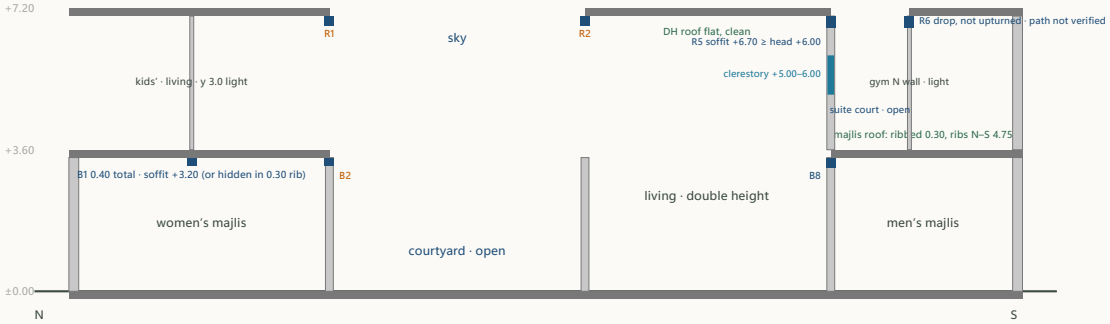
A · y 10: riwaq · courtyard · colonnade · kitchen



B · y 16.5: living (TV wall) · double-height void · dining · suite



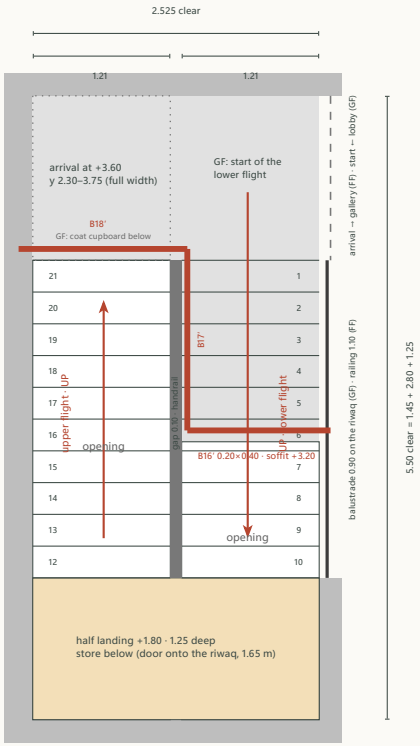
C · x 7.3: women's majlis · courtyard · double-height living · men's majlis roof · gym



The stair (flipped, option B · owner 2 Nov 2026) · plan at the first-floor opening and section · all sizes assumed

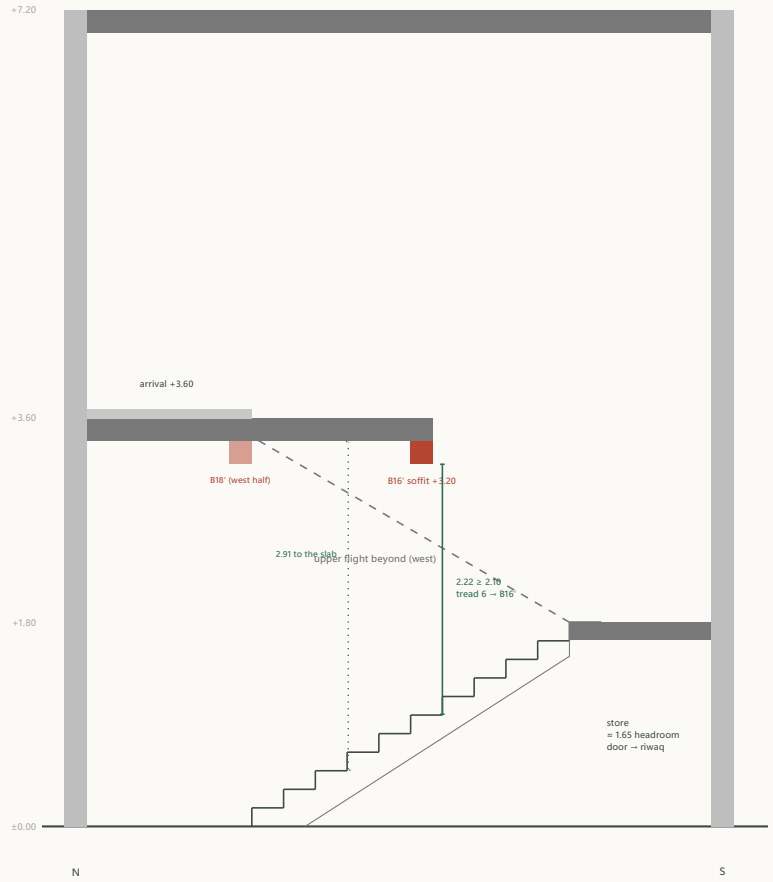
well x 0–2.75 · y 2.2–7.9 (wall centre lines) · 22 risers × 0.164 = 3.60 m · goings 0.28 · two flights side by side · the lower flight on the EAST half starts at the NORTH end off the family lobby, the half landing at the SOUTH end, the upper flight on the west half arrives at the NORTH end beside the lift door · headroom measured to the beam soffits

stair plan at the first-floor slab · white = opening · grey = slab · north up



section along the lower flight (x ≈ 2.0), looking west

V8.0: the roof slab over the well is closed (no R16–R18); V8.1: the roof is reached by the maintenance ladder in the suite court (X20)



Numbers (assumed)

- **Well:** 2.75 × 5.70 m on wall centre lines = **2.525 × 5.50 m clear** (0.25 outer wall west, 0.20 walls north and south; on the east the start opening (y 2.3–3.75) and a 0.90 balustrade along the lower flight (y 3.75–6.55) on the ground floor, a railing on the first floor; the store's wall y 6.55–7.9).
- **Rise:** 3.60 m = **22 risers × 0.164 m**, 11 per flight; **goings 0.28 m**, 10 per flight = 2.80 m.
- **Plan (flipped, option B):** the arrival / start strip 1.45 m (y 2.30–3.75) at the north end, flights y 3.75–6.55, the half landing 1.25 m (y 6.55–7.80) at the south end: **1.45 + 2.80 + 1.25 = 5.50 = the clear length. It fits.** The lower flight on the east half rises south, straight off the family lobby; the upper flight on the west half returns north and arrives beside the lift door. The west half of the start strip (1.21 × 1.45, full height) is the coat cupboard by the entrance.
- **Flight clear width:** (2.525 − 0.10 gap) / 2 = **1.21 m each**, ≈ 1.13 m between a wall handrail (0.08) and the central handrail; ≥ 0.90 m (residential).
- **Flight slabs:** 0.15 m solid, spanning landing to landing (2.80 m); the half landing on the stair walls (store under it, ≈ 1.65 m headroom: a store, not a room; its low door onto the riwaq by the courtyard glass).

Opening in the first-floor slab (L-shaped, ≈ 8.2 m²)

- **Over the lower flight only beyond tread 6:** the slab above a tread must leave ≥ 2.10 m to the **beam** soffit (+3.20 with 0.40 total trimmers), so treads with a level ≤ 1.10 m may stay under the slab: treads 1–6 (tread 6 at +0.98 → 2.22 m). The opening edge B16′ is at y 5.25 (beam 5.15–5.35), over tread 6's south part (y 5.15–5.43) — tread 7 (y 5.43–5.71, +1.15) is under open sky.
- **Over the upper flight and the half landing:** fully open (x 0.125–1.39, y 3.75–7.80 and x 1.39–2.65, y 5.35–7.80).
- **Trimmers, 0.20 × 0.40 total, soffit +3.20 (the same zigzag as V8.1, mirrored):** B16′ (y 5.25, from the riwaq line beam B10 to B17′), B17′ (x 1.49, on the lower flight's side of the split so it is never over the upper flight, from B16′ to B18′), B18′ (y 3.65, the arrival edge, from the west wall to B17′). B16′ and B18′ bear on B17′: one beam-on-beam zigzag, normal at a stair well; the engineer takes the torsion. **Option for the engineer:** 0.20 × **0.30** trimmers (soffit +3.30) would give 2.32 m at tread 6 if the 1.4–1.6 m spans allow it.
- **Headroom (over the full width of every tread):** 2.22 m at tread 6 to B16′ (the trimmer spans the whole east half); 2.91 m at tread 3 to the slab; under B17′ (over the lower flight's inner 0.10 strip, treads 1–5, level ≤ 0.82) ≥ 2.38 m; the upper flight and the half landing under the open well to the roof slab (≥ 3.4 m). **No conflict with the reservation.**
- **Arrival:** the upper flight lands at +3.60 on the arrival strip y 2.30–3.75 inside the stair box, which opens east onto the gallery's north part 0.10 m south of the lift notch; the former landing room (y 7.9–9.2) is the family bath's north part (F-b) behind a wall on B12.

Special zones · options compared (none imposed)

1 · The west projection: edge beam B3 on C21–C23 (spans 6.5 and 6.25 m)

	(a) drop beam	(b) upturned beam	(c) deeper slab edge
size (total, incl. the 0.20 slab)	0.25 × 0.50: 0.30 below the slab, soffit +3.10	0.25 × 0.50: 0.30 above the structural slab (top +3.90), i.e. ≈ 0.22–0.25 visible above a finished floor of 0.05–0.08; if the engineer needs 0.60 total, 0.40 above the slab	slab 0.35 over the 1.35 m strip, no beam
what you see	a 0.30 m downstand along the colonnade, 3.10 clear under it; the courtyard reads as a riwaq with a beam	a flush soffit at +3.40 over the colonnade; the upstand is the solid base of the gallery railing, the railing on it up to 1.10	flush soffit, a thicker slab edge 0.35 seen from the courtyard
structure	standard; the stiffest	standard; the slab hangs from the beam's bottom (stirrups) – routine	a 1.35 m wide, 0.35 deep band over 6.5 m: depth/span ≈ 1/18, deflection-sensitive, heavy
cost	lowest	≈ same formwork; saves the railing base	more concrete and steel
note	an acceptable alternative	**the owner's architectural preference**: flush colonnade soffit, railing base included; the engineer decides between (a) and (b)	only if a flush soffit AND an open railing are wanted

2 · The double-height living: a clean ceiling (preferred, not a rule)

- **Option 1 · flat solid roof slab 0.22 m spanning 4.7 m east–west** from the FF west rooms' wall (R3) to the suite wall (R4): no beam in the void, the two beams sit on the walls (invisible). A common span for a 0.22 solid slab (not calculated); cost ≈ a normal solid slab. **This study's preference.**
- **Option 2 · ribbed 0.30 slab, ribs east–west:** lighter and cheaper for the contractor, but a ribbed soffit (or a false ceiling) over the void; still no cross beam.
- **Option 3 · a cross beam on y 16.1 (mid-void) + 0.18 slab:** the cheapest in material, but a 0.45 m beam across the void, 7 m up, in the family's main room. Not recommended unless the engineer finds option 1 too deep.
- The void's walls: x 4.25 and x 8.95 are 7 m high but tied by the FF slab on their outer sides (the gallery / projection on the west, the suite floor on the east). The south wall (y 19.25, x 5.6–8.95, with the clerestory) is tied at +3.6 only by the men's majlis roof slab on its south face, and at +7.0 by the roof: **one tall wall, 7 m, tied on one side at mid-height** (X7).

3 · Living ↔ dining opening (x 8.95, 6.25 m) under the suite wall: beam B5

- A drop beam 0.25 × 0.55 total between C27 (8.95, 13.0) and C28 (8.95, 19.25); it carries the suite bedroom's west wall (3.4 m block), half the dining roof (ribbed, 4.36 m) and the roof beam R4 above the wall. 0.35 below the 0.20 slab: soffit +3.05 before finishes.
- **It can sit exactly on the ceiling-height change line:** the dining ceiling is at +3.40, the living has no ceiling, so the beam's 0.35 m below the dining ceiling becomes the step itself: one face of the double height. Headroom under it 3.05 m (≥ 2.40 required).
- **Alternative:** a wide-shallow beam 0.60 × 0.40 total (soffit +3.20) if the step should be thinner; or make the suite wall a light partition so the beam carries only the slab (0.25 × 0.45 total, soffit +3.15).

4 · Men's majlis roof (10.56 × 4.75 m) carrying the gym, the arm and the office

- **Span direction north–south, 4.75 m:** a ribbed (hollow-block) slab 0.30 m (25+5), ribs from the majlis north wall (B8) to the south wall (B9). **No column inside the majlis** (the room must stay free: 10.56 m long, seating at the east end).
- **South wing v2 (1 Nov 2026), traced:** the **gym** (x 0–4.25) sits inside the column frame C6 / C13 / C19 (y 19.25) – C7 / C14 / C20 (y 24): its floor is the west wash block's 0.20 solid slab on the west outer wall and B10 (x 2.75, C13 – C14, over the foyer's 1.8 m opening) plus a 1.5 m strip of the majlis roof to B19; its east wall stands on **B19 between C19 and C20** (a frame line, no transfer); its north double wall on B8 (x 2.75–4.25) and on the new drop beam **B22** (C6 – C13: no ground-floor wall under x 0–1.25); its south wall on B9. The treadmill stands on the 2.75 m solid bay next to the outer wall (the stiffest bay of the wing).
- The **office** (x 10–13.311): its roof spans N–S 4.75 m onto R5 / R8 with an edge rib R19 on the court, so its west wall x 10.0 is **non-load-bearing** (own weight on rib B21). The **court** (x 4.25–10): a terrace on the majlis roof (ribbed 0.30, 4.75 m): a depressed zone 0.10–0.15 for the falls and the low thresholds, waterproofing, a box gutter along the south side, linear drains at the doors; live load as a private terrace (≈ 3 kN/m², SBC 301, the engineer), finishes ≈ 2 kN/m², planters local. The gym's **nib** (1.55 × 1.3): light partitions, its roof on R3 and a 1.3 m cantilever R21 from C23.
- **Removed:** the gym's south arm, its north partition (y 21.9), R6 and its two bearings; the treadmill arm. Across the court's south side **R8** (C20 – C29 – C36) carries the cornice and the parapet as a facade frame (the infill below is the owner's choice).
- **Loads:** it is a **home gym** (light equipment, no dropped heavy weights, one treadmill): the engineer chooses a normal residential-gym live load as they judge (SBC 301 gives 5 kN/m² for gymnasia; what applies to a home gym is the engineer's call); the treadmill ≈ 1.5–2 kN dynamic (manufacturer's data); office 2.5–3 kN/m². **Vibration: not calculated here.** The natural frequency of the 4.75 m ribbed slab and its response under the treadmill (now beside the stiff west edge, on B19) must be checked by the engineer; the treadmill gets a floating floor and pads (DETAILS.md) in any case (X6).

5 · The stair (x 0–2.75, y 2.2–7.9)

- Drawn on the stair page: two flights of 1.21 m clear, 22 risers of 0.164 m, goings 0.28; **flipped (option B, 2 Nov 2026)**: the half landing 1.25 m deep at the south end, the start and the arrival at the north end beside the lift; the L-shaped opening of about 8.2 m² and the trimmers B16'–B18' (0.40 total, soffit +3.20); headroom 2.22 m at the opening edge, measured to the trimmer soffit (0.30 trimmers would give 2.32 m: the engineer). **It fits the reservation.**

6 · Glazing and the grid

- **Courtyard glazing** (x 4.25 y 6.8–8.9 and 10.7–12.7; x 8.95 y 9.1–11.1): no column in the openings (C17, C18, C26, C27 at the corners); beams B4 / B6 above the heads. **Women's majlis south glazing** (y 6.5, x 5.1–8.1): C21 at x 5.8 sits in it → the glazing is 5.1–5.6 + 6.0–8.1 (accepted with the colonnade; V6.4). **Living glazing** (y 13.0, x 4.45–8.75): C22 at x 5.8 is the pier (the columns solution, tag v5.0). **Clerestory** (y 19.25, x 6.0–7.0 and 7.6–8.6, +5.00–6.00): C23 at x 5.6 and C28 at x 8.95 are clear of it; the roof beam R5 soffit +6.70 is 0.70 above the head. **Suite bath east window**, shortened by the owner to y 9.6–10.2: the clearance to C33 depends on the column's length along the wall, see X3.

7 · Lift shaft, air tower, roof openings

- **Lift (future, two stops):** RC walls W1 on x 0, y 0 (outer) and y 2.2, 0.20 m; the east wall x 2.25 blockwork with the future door reveal. The FF slab and the roof over the shaft (2.25 × 2.2) are **framed as openings now and closed with removable precast / lightweight panels** on hidden beams (B11 and the x 2.25 line), so the shaft opens later without cutting. Pit 1.2 m, isolated from the air tunnel.
- **Air tower** (8.95–10.731 × 0–1.0): rises through both slabs; the openings framed by B15 / R9 (y 1.0) and the walls x 8.95 (W3) and the cut wall (W4); the head 2.5 m above the roof in the same RC walls (studies/air_system.typ). **Below the slab the shaft walls continue to −1.95 as a dry base pit**, from which the small air duct (a sealed box 0.9 × 0.9, top −1.00, underside −1.90) leaves westward through the W3 line and runs south under the women's majlis on x 7.27 to the outlet chamber under the courtyard floor (x 6.75–7.95, y 6.6–7.6, floor −2.10, between the C21 and C26 pads); the outlet is a grille in the wall base at +0.2 … +0.8 through a plenum in the wall. The pit is the foundation of C24 / C30 and of the W3 / W4 ends; the women's majlis south wall bridges the duct mouth on a ground beam (air study items A1–A3). No water anywhere in the system.
- **Roof openings:** courtyard sky framed by R1, R2, R11 (west, on the colonnade, X8) and R4, the gallery strip roof between R3 and R11; suite court by R5, R6 and the FF walls x 5.6 / x 11.586 below the roof. West shaft (1.5 × 1.2) framed in B13 / R10.

Conflicts and open questions for the structural engineer

No.	State	Conflict / question
X1	accepted	Column **C22** (5.8, 13.0; V6.4, was 5.6) stands in the courtyard / living glazing (x 4.45–8.75): the glass is split 4.45–5.55 + 6.05–8.75 (the Ø0.40 stone sleeve); above the glass head the steel window W3 (+2.90 … +6.75) closes the void to the courtyard (V6.4, enclosure A). A decision of the columns solution (tag v5.0); keep the column round Ø0.30 and detail the glazing around it.
X2	accepted	Beam **B5** (x 8.95, 6.25 m, 0.25 × 0.55 total) shows 0.35 m below the dining ceiling, on the ceiling-height change line, soffit +3.05 before finishes. Acceptable: it is the step between the two heights. Alternative 0.60 × 0.40 total (soffit +3.20).
X3	owner decided · resolved (V6.3)	**C33 stays at (13.311, 10.45); the suite bath east window is 0.55 m, y 9.50–10.05 (V6.3).** Its frame (0.05) clears the assumed 0.25 × 0.60 column lying along the wall (y 10.15–10.75) by 0.05 m, and any smaller column by more: no clash for any column ≤ 0.60 along the wall. The GF kitchen window (y 8.7–10.1) is clear.
X4	superseded (v2)	R6 is removed (south wing v2): the court is open from the living's clerestory windows to the south facade (4.75 m); nothing stands in front of the windows but the facade frame (parapet +8.40 at y 24). The winter noon sun reaches the whole window height (studies/south_wing.typ § v2).
X5	load path revised (v2, 1 Nov 2026), NOT verified	**Load path revised, not verified · traced: it IMPROVED.** V7.1 carried R6 (6.0 m) on two first-floor walls standing on ribs of the men's majlis roof (a double transfer); v1 shortened it to 4.4 m. **v2 removes R6:** the court is open to the south facade. The gym's east wall (x 4.25) now stands on **B19 between the existing columns C19 and C20** (a frame line, not a transfer); the office roof spans N–S onto R5 / R8, so the office's west wall (x 10.0) carries only its own weight on rib B21; the double wall to bedroom 3 is on B8 and the new drop beam B22 (C6 – C13). What remains on the majlis roof: the court terrace (finishes, falls, planters, people), the nib's light partitions and the x 10.0 wall's own weight. **The engineer must confirm:** B19 (hidden 0.60 × 0.30 or drop 0.25 × 0.50) and B22; R8 across the court as the facade frame (4.7 + 1.05 m, the cornice and the parapet on it); the court terrace load and the depressed zone; the treadmill's bay (the 0.20 solid slab, 2.75 m) and its vibration (floating floor and pads, X6); the nib roof cantilever R21; the light-roof option. In the 3D model the x 10.0 wall stays **unconfirmed** (red) until then. Loads: a home gym, see § 4.
X6	flag	The gym's treadmill (v2) on the west wash block's 0.20 solid slab (2.75 m span, the outer wall and B10): a floating floor or dense rubber and anti-vibration pads; the slab's response under it is the engineer's. The nib's south partition (y 20.55, 1.55 m) crosses the majlis ribs: a light partition on the slab.
X7	flag	The living's south wall (y 19.25, x 5.6–8.95) is 7 m tall with the clerestory windows, tied at +3.6 only on its south face (men's majlis roof slab) and at +7.0 by the roof: its slenderness between columns C23 and C28 (3.35 m) is **not checked here**; the engineer confirms it, with RC stiffeners at the window jambs if needed.
X8	owner decided	**Colonnade columns C21–C23 continue to the roof** inside the gallery railing line (a two-storey colonnade, slender round columns ≈ Ø0.30 assumed). The roof over the gallery strip (1.55 m, y 6.5–13.0; V6.4) rests on a roof edge beam **R11** (x 5.8, 0.25 × 0.45 total) between C21 and C22: no roof cantilever. C22 also takes the west end of R2, C23 stands under R5. The engineer sizes the columns for two storeys and the two roof beams.
X9	missing	**No soil report, no soil data:** footing type and depth (pads ≈ 1.5 × 1.5 assumed, underside ≈ −1.50), the lift pit (1.2 m), and the **air duct** (studies/air_system.typ v2): a small sealed box 0.9 × 0.9, top −1.00 / underside −1.90, leaving the tower base pit (to −1.95) through the W3 / C24 line and running under the women's majlis to the outlet chamber under the courtyard floor (floor −2.10) between the C21 and C26 pads; the pit as the foundation of C24 / C30 and the W3 / W4 ends, the ground beam of the majlis south wall over the duct mouth: engineer (air items A1–A3).
X10	flag	**Suite walls over the kitchen** (x 10.3, y 10.45, x 12.211, y 8.0, y 11.9, y 13.9): light partitions on the ribbed slab, or hidden beams under the bath walls that carry pipes; the **suite bath floor is sunken** (depressed slab ≈ 0.15 over x 10.3–13.311, y 6.5–10.45) with nothing in the kitchen ceiling: a depressed-slab detail inside a ribbed bay.
X11	flag	**Floor-to-floor 3.60** with a 0.20 slab and beams of 0.40–0.55 total depth gives 3.05–3.20 m clear under the beams before finishes (B4 soffit +3.10 over the riwaq glazing, B5 +3.05 at the dining step, B1 +3.20 in the women's majlis). Acceptable (≥ 2.40). **Owner decision:** B1 is a drop beam hidden by a gypsum ceiling (≈ +3.05); the general rule is that drop beams in finished rooms are hidden by gypsum ceilings, which also carry the hidden AC and lighting.
X12	open	**Beam B4** (x 4.25, 6.5 + 6.25 m) carries the first-floor bedrooms' east wall over the riwaq glazing (y 6.8–8.9, 10.7–12.7) and the low wall: with 0.25 × 0.50 total (soffit +3.10) the riwaq glazing heads are at ≤ +3.10 before finishes, and the dining-side heads under B5 at ≤ +3.05; confirm the window heights with the architect.
X14	owner decided (V6.4)	**The kitchen's north wall moves to y 7.0** (maid room to the municipal minimum): a **non-load-bearing block partition** on a thickened slab / ground beam (it crosses the malqaf duct at x 10.25 on a ground beam, air study A3). The structural grid is unchanged: the y 6.5 line keeps **B6** as a **drop beam** (0.25 × 0.45 total assumed) from C26 to the east wall, inside the laundry / maid room ceiling behind the gypsum ceiling, carrying the first-floor bedroom 1 / suite wall; the FF slab bays stay on the grid. W5 is dropped from the RC-wall proposal (X13).
X15	owner decided (V6.4)	**Courtyard enclosure, option A** (studies/courtyard_enclosure.typ): the west projection is **1.55 m** (edge x 5.8); B3, R11 and C21–C23 move 0.20 m east, spans and X8 unchanged. The gallery edge, the family living's edge and the living's upper courtyard side (y 13.0, +2.90 … +6.75) are closed with black steel windows W1 / W2 / W3; W1 sits on the B3 upstand (its sill), heads at the R1 / R2 / R11 soffit +6.75; a 0.25 m fascia keeps the head line under the roof slab over the void. W3 needs a steel transom at +3.82 tied to C22 and the kitchen wall; the window loads are small. The gym route wall (B20) stays on x 5.6: a 0.20 m nib at y 19.25 (to follow to x 5.8 in the L study if the owner wants one line).
X16	superseded (1 Nov 2026)	**The main stair STOPS at the first floor** (owner, studies/south_wing.typ): flights 3–4, the roof stair room and the roof opening over the well are dropped, the roof slab over the well is an ordinary bay (no R16–R18); the roof is reached by a **maintenance ladder in the suite court** (V8.1, X20, studies/roof.typ page 2).
X17	owner decided (V6.4)	**Roofline (studies/roof.typ):** flat roof, no tiles. A **main honey-stone cornice** 0.45 × 0.30 projection at +6.90 … +7.35 all round on the slab edge (≈ 2.5 kN/m, anchored); on the north block (y 0–19.25) a 0.30 upstand with coping to +7.65 and an **attic screen wall** (200 block; V8.1 **lowered**: 1.3 m above the slab = 1.40 above the finish, coping +8.70; was 1.8 / +9.10) set back 1.0 m from the facade face, i.e. 0.875 m inside the outer-wall beams: a line load ≈ 3.6 kN/m on the slab near its supports (thicken the slab edge or a small hidden beam); its south wall on R5 / B8. The **south wing** (y 19.25–24): V8.1 a **low parapet to +8.00** (0.70 above the finish, ≈ 1.8 kN/m) and a **continuous 1.10 m black steel guardrail set back 1.0 m** on the slab (posts at 1.5 m on base plates through the waterproofing or on a 0.20 kerb; 0.74 kN/m horizontal at the top rail and a 0.9 kN point load, SBC 301 / ASCE 7 guards: the anchors and the upstand detail are the engineer's); the strip outside the rail a no-go zone. **Plant loads** (reserve): AC units ≈ 1.0 kN each + ERV 1.5 kN on pads over the maid wing / suite bath roof; tank(s) up to 30 kN on ≈ 3.3 m² in the R10–R3 bay (x 2.85–4.15, y 1.0–3.5); PV ≈ 0.25 kN/m² + wind uplift on the south wing (y ≥ 22.0) and the north block east roof; maintenance live load 1.0–1.5 kN/m². All assumed.

X18	owner decided (V7.1)	**Courtyard arcade and arched enclosure windows** (studies/courtyard_enclosure.typ): two slender stone columns Ø0.25 on the colonnade line x 5.80 at y 8.667 and 10.833 (courtyard bay only) with three semicircular arch plates under the projection's soffit (+3.27; springing ≥ +2.31): **decorative, carrying nothing in this study**; the engineer may use them as intermediate supports of B3 (spans 2.17 m instead of 6.5 m) if that helps, in which case they get footings. The enclosure windows W1 / W2 / W3 get semicircular heads with their crowns at +6.60, 0.15 m below the R11 / R2 soffit: the wall between the windows and above the arches is 200 mm block on the B3 upstand (W1) or on B2 (W2); W3's head band 0.15 m under R2. No change to the beams or the columns.
X19	owner decided (air v3, 1 Nov 2026)	**Air system v3 (studies/air_system.typ), notes and loads only, all assumed:** (1) the **breeze channel** in the ground slab, 0.9 wide × 0.20 clear within the slab depth (−0.08 … −0.28) on x 10.2 from the tower to y 10.5 then west to the kitchen's west wall, with its own 0.10 base slab and removable covers flush with the floor in the laundry and kitchen walkways; through the walls y 1.0 (the tower's south wall), y 2.8 and y 7.0 a 0.9 × 0.2 slot in the wall foot **above** the ground beams' top (−0.28): no beam or footing notched; the throat under the maid bath 0.5 wide; a sump 0.4 deep at the tower's foot, no pit. (2) The **tower**: an airtight partition (100 mm) along its length, the head divided by a thin + divider over 1.2 m, a roof hatch 0.6 × 0.8 in its south face; two Ø250 risers and a Ø315 duct through its south wall at the roof. (3) The **HRV base under the tower** (v3.1, owner): the tower rises from a base y 1.0–3.4 on the tower's west line and the cut facade line (≈ 5.5 m² gross), its walls (0.25 on the cut, 0.20 elsewhere) 1.8 m high to the top +9.10 ≈ 5 kN/m on the roof slab over the maid bath / bedroom 1 bay and on the cut wall line, a 0.15 slab, the unit ≈ 1.1 kN on a plinth (≈ 1.5 kN/m² local), roof penetrations for the risers inside the tower's compartment; doors on its west face, the exhaust louvre in its cut face. (4) The chamber, the pit and the walk-in tunnel of v1 / v2 are withdrawn: no excavation below the slab except the channel's base and the sump.
X20	V8.1 · coordination	**The maintenance ladder's recess (studies/roof.typ page 2):** a honey-stone **pilaster 1.30 × 0.35 added on the court face of the gym's east wall** (x 4.35–4.70, y 20.65–21.95) holds the 0.90 × 0.30 recess for the ladder; the **0.20 wall on B19 (under R3) is not cut**. The pilaster (≈ 1.7 m³ block ≈ 20 kN) stands on the court's depressed slab (the men's majlis roof): a local load for the engineer. **Alternative:** if the engineer confirms the x 4.25 wall as non-load-bearing infill between B19 and R3, the recess may be cut into a locally rebuilt 0.50 wall (the pilaster disappears). The ladder's fixings (≈ 1.5 kN each) into solid block or a cast-in steel sub-frame; the gate posts of the court guard on R3; the stiles 1.10 above the roof. The pilaster's foundation is the slab: **no footing**.
X21	3 Nov 2026 · note	**First-floor wet rooms by the west shaft (owner, 3 Nov 2026):** the F-b bath is split into bedroom 2's en-suite (y 9.2–11.0, with the former niche) and a family WC (y 7.9–9.2): a **0.20 service wall on y 9.2 from x 1.5 to 4.25** (the shaft's north wall line continued; light steel-frame wall carrying two wall-hung WCs, a basin and the shower channel's trap, ≈ 1 kN/m) on the solid 0.20 west-band slab; no new opening in the slab, every pipe in the wall into the shaft. **Ground floor:** the family basins and WC become one powder room 2.75 × 3.2: the wall on y 14.4 is removed, so the hidden beam **B14** either becomes a drop beam or the bay spans 3.2 m (the engineer); a 0.25 service wall on y 16.1 (the serving passage side) holds the WC frame and bedroom 3's stack. No ground-floor shower (no sunken zone needed).
X13	open	**Which walls are RC:** this study proposes W1–W4 (lift, shaft, laundry wall, cut wall) as RC 0.20 m for stiffness (W5, the kitchen north wall, dropped in V6.4), plus the two gym walls of X5. The engineer confirms the lateral system (frame + walls) under SBC 301 and the wind / seismic case for Taif.
X14	open	**Slab types:** ribbed 0.30 (hollow block) for the 4.36–4.75 m bays and the men's majlis roof; solid 0.20 elsewhere and 0.22 flat over the double-height living. The engineer may unify (all ribbed, or all solid 0.20–0.25) for buildability; the clean soffit over the void is the only architectural wish.
X15	open	**Stair** (drawn on the stair page; flipped 2 Nov 2026, option B): flight widths 1.21 m clear, 22 risers × 0.164, the L-shaped opening turned round and the trimmers B16′–B18′ with their torsion (B18′ on the west wall, B16′ on B10), the store under the half landing at the south end (≈ 1.65 m headroom), handrails; the arrival strip y 2.30–3.75 at the first floor; a wall on B12 (y 7.9) closes the opening's south side (the family bath's north part).

Rules of thumb used (all assumed): beam depth = TOTAL depth incl. the slab, drop beams ≈ span / 12, hidden beams = slab depth, solid one-way slabs ≈ span / 24–28 (0.20 for ≤ 5 m), hollow-block ribbed slabs 0.30 (25 + 5) for 4–6 m, flat solid roof 0.22 for 4.7 m, columns 0.25 × 0.50–0.60 in the walls (two storeys, light loads), colonnade Ø0.30, pad footings 1.5 × 1.5 m. Loads: SBC 301 (residential 2 kN/m², gym: a home-gym live load as the engineer judges, roof 1 kN/m² + plant). Nothing here replaces the structural design.

✓ **Preliminary structural study · 36 columns · 22 first-floor beams · 13 roof beams · 23 items for the engineer**